
```

def getPermsWithRep(chars, permLength=None, prefix=''):
    indent = '.' * len(prefix)
    print(indent + 'Start, args=(' + chars + ', ' + str(permLength) + ', ' + prefix + ')')
    if permLength is None:
        permLength = len(chars)

    # BASE CASE
    if (permLength == 0): ❶
        print(indent + 'Base case reached, returning', [prefix])
        return [prefix]

    # RECURSIVE CASE
    # Create a new prefix by adding each character to the current prefix.
    results = []
    print(indent + 'Adding each char to prefix "' + prefix + '".')
    for char in chars:
        newPrefix = prefix + char ❷

        # Decrease permLength by one because we added one character to the prefix.
        results.extend(getPermsWithRep (chars, permLength - 1, newPrefix)) ❸
    print(indent + 'Returning', results)
    return results

print('All permutations with repetition of JPB123:')
print(getPermsWithRep('JPB123', 4))

```

The equivalent JavaScript program is in *permutationsWithRepetition.html*:

```

<script type="text/javascript">
function getPermsWithRep(chars, permLength, prefix) {
    if (permLength === undefined) {
        permLength = chars.length;
    }
    if (prefix === undefined) {
        prefix = "";
    }
    let indent = ".".repeat(prefix.length);
    document.write(indent + "Start, args=(\"" + chars + "\", " + permLength + ", \"" + prefix + "\")<br />");

```